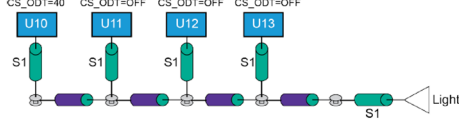
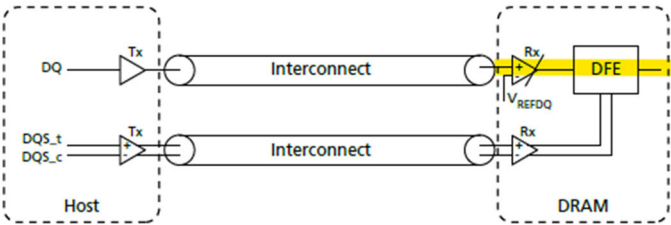


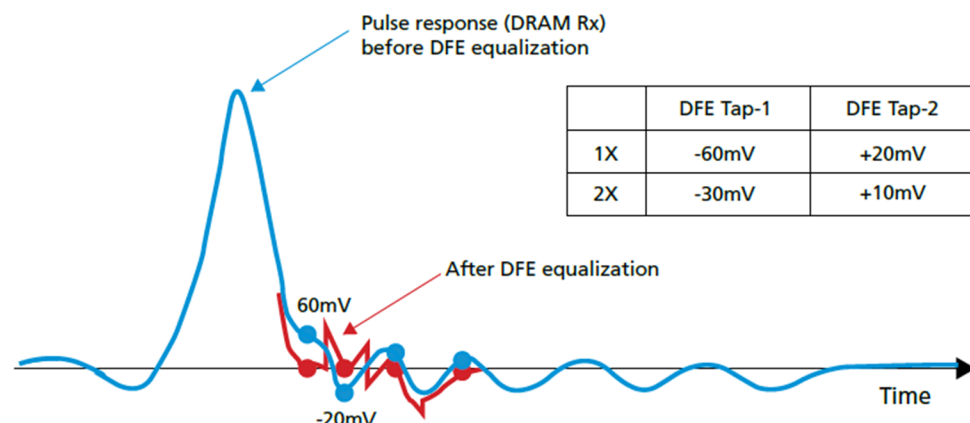
U.S. Patent No. 10,025,731 Infringement Claim Chart

Claim Language	'731 Accused Products ⁴
	DDR5 On-Die Termination DDR5 module designs incorporate the same basic routing topologies for all I/O, address, control/command and clock signals as DDR4: • The familiar input/output (DQ) and input/output strobe (DQS) pins are all direct-routed from the edge connector or data buffer. • Clock, command and address pins are fly-by routed. The difference is with the termination method: DDR5 uses on-die termination (ODT) instead of discrete termination used on DDR4. • DDR4 uses discrete termination resistors on the modules/boards for command clock (CK), chip select (CS), command/address (CA) and other control pins. • DDR5 adds the benefit of programmable ODT for CK, CS and CA, as well as a per-device configurable CA_ODT pin. The CA_ODT pin is a new feature on each DDR5 SDRAM device that enables the last DRAM on a CS, CA, or CK net to have a comparatively strong ODT setting (40/80 ohms), while all remaining DRAM on the CS, CA, or CK net have weak or disabled (ODT) settings (disabled/480 ohms) (Figure 5). The CA_ODT pin can be tied/strapped HIGH or LOW on the board/DIMM to enable one of two different groupings, which have mode register-programmable ODT levels. Typical DIMM or board usage applies a weak termination setting to Group A (devices nearer to the controller) and a stronger termination setting to the Group B (device at the end of the fly-by routing). DDR5 CS_n Fly-by Routing and ODT Setting Example  Figure 5: DDR5 Additional Routing Technologies Ex. CC-11 (“DDR5 Features White Paper”) at 4. The DFE circuits (see limitation [1.d] above) make signal corrections by equalizing the DQ signals without amplifying noise due to insertion loss and reflections:

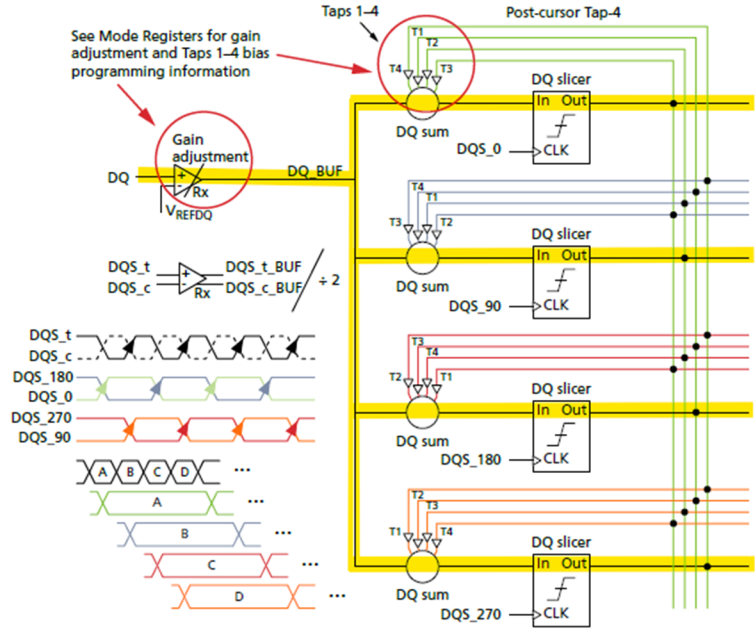
U.S. Patent No. 10,025,731 Infringement Claim Chart

Claim Language	'731 Accused Products ⁴
	At data rates ≥ 2933 MT/s, signal degradation due to inter symbol interference (ISI) is expected to increase and the data eye at the device ball is expected to be closed. Because the memory channel is very reflective due to the many impedance mismatched points that exist along the memory subsystem, ISI due to reflections is expected to increase. Traditional methods of characterizing the receiver using the input eye mask are no longer sufficient. DDR5 requires equalization to help improve (or open up) the data eyes after data is latched by the receiver. A 4-tap decision feedback equalization (DFE) helps equalize the DQ signals without amplifying the noise due to insertion loss and reflections, which is a common side effect of other equalization techniques (for example, CTLE). The following figure shows an example of a memory subsystem with DFE circuit included on the device. At the 1980-2100 MT/s data rates, the device's DFE is disabled. Setting either the global DFE gain and tap 1-4 enable bits to disable or setting all DFE gain and tap 1-4 bias bits to step 0 disables the DFE. Ex. CC-04 (DDR5 Datasheet) at 263. Figure 156: Memory Subsystem with DFE Circuit on the Device  Ex. CC-04 (DDR5 Datasheet) at 263 (highlight indicating the signal path for signals transmitted along data path DQ).

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Claim Language	'731 Accused Products ⁴									
	Notes: 1. As speed increases, the impact of loss from the channel makes the bias range of the first cursor asymmetric. 2. Values are defined for the entire voltage, temperature range, and Rx V_{REF} range from 0.5 * V_{DDQ} to 0.9 * V_{DDQ}. 3. Values are identical for taps 1-4. 4. These parameters are suggested to evaluate relative ratio of DFE taps 1~4, and absolute values of all parameters are not subject to silicon validation nor production test. 5. For the pulse response (...000010000... pulse pattern), a positive value corrects a negative post-cursor by setting the DFE tab bias sign bit (MR113~116, OP[6]) to 0 to apply a positive correction. 6. For the pulse response (...000010000... pulse pattern), a negative value corrects a positive post-cursor by setting the DFE tab bias sign bit (MR113~116, OP[6]) to 1 to apply a negative correction. For example, in a memory channel where the ISI during the first post-cursor is dominated by bandwidth loss, the expected tap-1 bias sign bit will be set to 1. 7. Check the multiplier that is applied to the DFE tap bias setting (MR113, MR114, MR115, etc.) for total DFE feedback swing implemented in hardware. The figure below is an example of 1X and 2X multiplier cases which shows how the DFE tab biases move differently in hardware with the same pulse response. Figure 159: Example Case of DFE Tap Biases Moving with 1X and 2X Multiplier <table data-bbox="1310 875 1717 1005"><thead><tr><th></th><th>DFE Tap-1</th><th>DFE Tap-2</th></tr></thead><tbody><tr><td>1X</td><td>-60mV</td><td>+20mV</td></tr><tr><td>2X</td><td>-30mV</td><td>+10mV</td></tr></tbody></table> Ex. CC-04 (DDR5 Datasheet) at 265.		DFE Tap-1	DFE Tap-2	1X	-60mV	+20mV	2X	-30mV	+10mV
	DFE Tap-1	DFE Tap-2								
1X	-60mV	+20mV								
2X	-30mV	+10mV								

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	Figure 162: 4-Way Interleave 4-Tap DFE  Ex. CC-04 (DDR5 Datasheet) at 267 (highlight indicating the signal path along DQ lines). Impressively, this dramatic increase in the I/O switching rate (data rate) is achieved without the need for differential signaling at the DQ pins; the DQ bus remains single-ended, pseudo-open drain (POD). However, there are critical new features that enable these higher data rates to be achieved. One of these is the addition of equalization in the form of a multi-tap decision feedback equalizer (DFE) in the DQ receivers. The DFE mitigates the effects of inter-symbol interference (ISI) at the higher rates by opening up the data eyes inside the device. Ex. CC-08 (DDR5 White Paper) at 3.

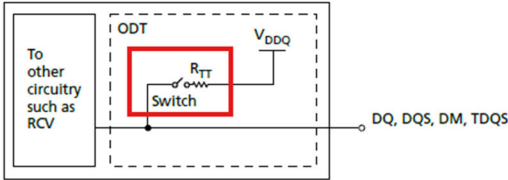
U.S. Patent No. 10,025,731 Infringement Claim Chart

Claim Language	'731 Accused Products ⁴
	DQ and DQS Training DQ and DQS training is done in a multiphase set of routines using newly defined features and functions. These individual routines, such as read training, read preamble training and write level training, align DQS pins to DQ pins and center-align these to a reference clock. The Duty Cycle Adjuster (DCA) feature can be used to adjust for improved alignment between the DQS and DQ internal signals. The DQ and DQS transmitters (Tx) / receivers (Rx) are then trained using new multitap decision feedback equalization (DFE) settings to adjust for crosstalk and noise on each subchannel and improve data eye width. VrefDQ global and per-bit training is used to adjust the signal for data eye height. To combat the effects of voltage and temperature shifts on the DRAM, a DQS interval oscillator helps signal to the system when it may need to retrain. These methods and new features are used to accommodate the stringent bit error rate (BER) eye metric required as data rates increase. Ex. CC-11 (DDR5 Features White Paper) at 5.
[1.f] the each correction circuit of the set of correction circuits including at least one programmable impedance matching circuit; and	In the '731 Accused Products each correction circuit of the set of correction circuits includes at least one programmable impedance matching circuit. For example, each correction circuit of the set of correction circuits includes an ODT circuit and DFE circuit, and the ODT circuit includes a programmable impedance matching circuit:

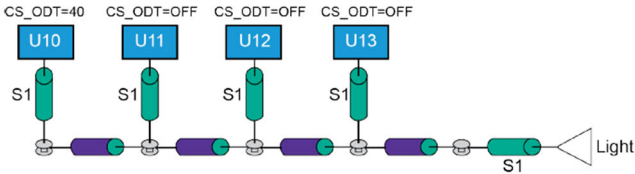
U.S. Patent No. 10,025,731 Infringement Claim Chart

Claim Language	'731 Accused Products ⁴
	On-Die Termination On-Die Termination for DQ, DQs, DM and TDS On-die termination (ODT) enables the DRAM to change termination resistance for each DQ, DQS, DM and TDQS pin. Unlike previous DDR technologies, DDR5 no longer has a physical ODT pin, and all ODT-based control is now command- and mode register-based. For x4 and x8 configuration devices, ODT is applied to each DQ, DQS_t, DQS_c and DM_n, as well as TDQS_t, TDQS_c for x8 configuration). For x16 configuration devices, ODT is applied to each DQU, DQL, DQSU_t, DQSU_c, DQSL_t, DQSL_c, DMU_n and DML_n. ODT is enabled via non-target READ/MODE REGISTER READ commands, WRITE and NON-TARGET WRITE commands, or is set to the default PARK value through settings in MR33-MR35. The ODT feature is designed to improve signal integrity of the memory channel by enabling the device controller to independently change termination resistance for any or all devices. In addition to the control capability of the DQ ODT, the DQS ODT is now independently programmed via MR33:OP[5:3] and held static. All ODT control is targeted for the DQs. This addition enables adjusting the delay common in an unmatched architecture. DQS RTT offset control mode is enabled via MR40:OP[2:0]. The ODT feature is turned off and not supported in self refresh mode, but does have an optional mode when in power-down. The figure below shows a simple functional representation of the ODT feature. Ex. CC-04 (DDR5 Datasheet) at 303.

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Claim Language	'731 Accused Products ⁴
	Enhancing data reliability and performance: Exploring On-die termination (ODT) in DDR5 memory Signal integrity is more important as data is delivered at faster speeds in DDR5 memory. When there is an imbalance between the characteristic impedance of the transmission line and the impedance of the connected devices, signal reflections may happen. DDR5 (Double Data Rate 5) memory modules and other high-speed digital systems use the on-die termination (ODT) technology to lessen signal reflections and enhance signal integrity. By placing a termination resistor that matches the transmission line's impedance right on the memory chip, on-die termination minimizes the possibility of signal reflections. Therefore, ODT is a crucial component for high-speed DDR5 memory systems since it aids in enhancing signal quality, decreasing signal ringing, and eventually allowing for higher data transfer speeds with less signal deterioration. Ex. CC-10 Kankana Mazumdar, <i>DDR5's Secret Weapon: On-Die Termination (ODT) for Noise Reduction and Power Efficiency</i>, Intellectual Partners (Nov. 24, 2023) at 2. Figure 192: Functional Representation of ODT  The switch is enabled by the internal ODT control logic, which uses command decode, mode register settings and other control information (see below). The value of R_{TT} is determined by the settings of mode register bits.

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	Ex. CC-04 (DDR5 Datasheet) at 303 (red box shows the programmable switch that dynamically places a termination resistor for impedance matching). The programmable ODT further includes a “per device configurable CA_ODT pin” for impedance matching. DDR5 On-Die Termination DDR5 module designs incorporate the same basic routing topologies for all I/O, address, control/command and clock signals as DDR4: • The familiar input/output (DQ) and input/output strobe (DQS) pins are all direct-routed from the edge connector or data buffer. • Clock, command and address pins are fly-by routed. The difference is with the termination method: DDR5 uses on-die termination (ODT) instead of discrete termination used on DDR4. • DDR4 uses discrete termination resistors on the modules/boards for command clock (CK), chip select (CS), command/address (CA) and other control pins. • DDR5 adds the benefit of programmable ODT for CK, CS and CA, as well as a per-device configurable CA_ODT pin. The CA_ODT pin is a new feature on each DDR5 SDRAM device that enables the last DRAM on a CS, CA, or CK net to have a comparatively strong ODT setting (40/80 ohms), while all remaining DRAM on the CS, CA, or CK net have weak or disabled (ODT) settings (disabled/480 ohms) (Figure 5). The CA_ODT pin can be tied/strapped HIGH or LOW on the board/DIMM to enable one of two different groupings, which have mode register-programmable ODT levels. Typical DIMM or board usage applies a weak termination setting to Group A (devices nearer to the controller) and a stronger termination setting to the Group B (device at the end of the fly-by routing). DDR5 CS_n Fly-by Routing and ODT Setting Example  Figure 5: DDR5 Additional Routing Technologies

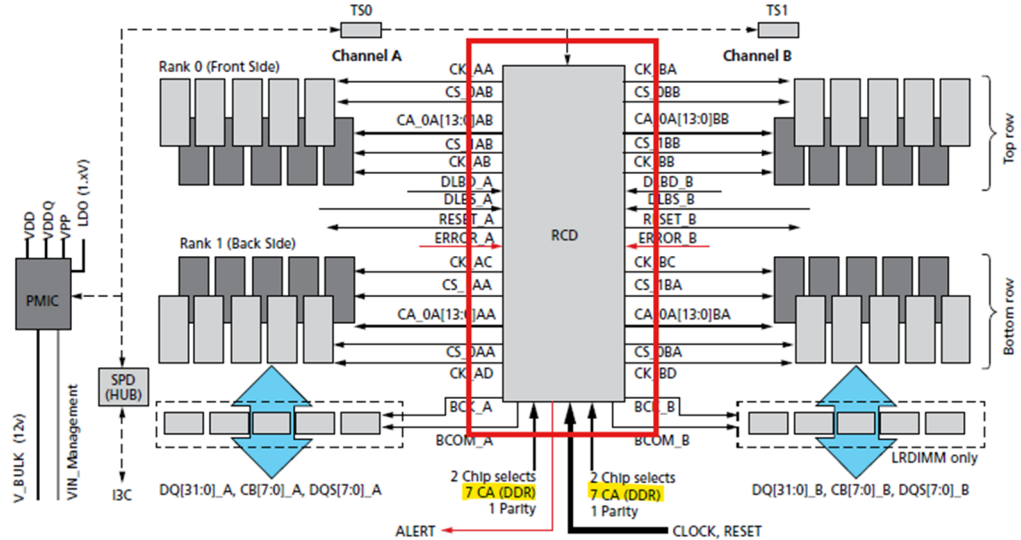
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	Ex. CC-11 (DDR5 Features White Paper) at 4. Table 182 — ODT Electrical Characteristics RZQ=240 Ω +/-1% Entire Temperature Operation Range; after Proper ZQ Calibration <table><tr><th>RTT</th><th>Vout</th><th>Min</th><th>Nom</th><th>Max</th><th>Unit</th><th>Notes</th></tr><tr><td rowspan="3">240 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ</td><td>1, 2, 3</td></tr><tr><td rowspan="3">120 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/2</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/2</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/2</td><td>1, 2, 3</td></tr><tr><td rowspan="3">80 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/3</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/3</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/3</td><td>1, 2, 3</td></tr><tr><td rowspan="3">60 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/4</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/4</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/4</td><td>1, 2, 3</td></tr><tr><td rowspan="3">48 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/5</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/5</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/5</td><td>1, 2, 3</td></tr><tr><td rowspan="3">40 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/6</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/6</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/6</td><td>1, 2, 3</td></tr><tr><td rowspan="3">34 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/7</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/7</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/7</td><td>1, 2, 3</td></tr><tr><td>DQ-DQ Mismatch within byte</td><td>VOMdc = 0.8* VDDQ</td><td>0</td><td>-</td><td>8</td><td>%</td><td>1, 2, 4, 5, 6</td></tr></table> NOTE 1 The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see "ZQ Calibration Commands" section. NOTE 2 Pull-up ODT resistors are recommended to be calibrated at 0.8*VDDQ. Other calibration schemes may be used to achieve the linearity spec shown above, e.g. calibration at 0.5*VDDQ and 0.95*VDDQ. NOTE 3 The tolerance limits are specified under the condition that VDDQ=VDD. NOTE 4 DQ to DQ mismatch within byte variation for a given component including DQS_T and DQS_C (characterized) NOTE 5 RTT variance range ratio to RTT Nominal value in a given component, including DQS_t and DQS_c. $\text{DQ-DQ Mismatch in a Device} = \frac{\text{RTTMax} - \text{RTTMin}}{\text{RTTNOM}} * 100$ NOTE 6 This parameter of x16 device is specified for Upper byte and Lower byte. Ex. CC-05 (JESD79-5) at 315 (showing the electrical characteristics for the resistance values that can be selected for a termination (RTT) for impedance matching); <i>see also</i> Ex. CC-04 (DDR5 Datasheet) at 304 (same). Moreover, the DFE circuitry mitigates issues generated from impedance mismatches:	RTT	Vout	Min	Nom	Max	Unit	Notes	240 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ	1, 2, 3	120 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/2	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/2	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/2	1, 2, 3	80 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/3	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/3	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/3	1, 2, 3	60 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/4	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/4	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/4	1, 2, 3	48 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/5	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/5	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/5	1, 2, 3	40 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/6	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/6	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/6	1, 2, 3	34 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/7	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/7	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/7	1, 2, 3	DQ-DQ Mismatch within byte	VOMdc = 0.8* VDDQ	0	-	8	%	1, 2, 4, 5, 6
RTT	Vout	Min	Nom	Max	Unit	Notes																																																																																																																																														
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	At data rates ≥ 2933 MT/s, signal degradation due to inter symbol interference (ISI) is expected to increase and the data eye at the device ball is expected to be closed. Because the memory channel is very reflective due to the many impedance mismatched points that exist along the memory subsystem, ISI due to reflections is expected to increase. Traditional methods of characterizing the receiver using the input eye mask are no longer sufficient. DDR5 requires equalization to help improve (or open up) the data eyes after data is latched by the receiver. A 4-tap decision feedback equalization (DFE) helps equalize the DQ signals without amplifying the noise due to insertion loss and reflections, which is a common side effect of other equalization techniques (for example, CTLE). The following figure shows an example of a memory subsystem with DFE circuit included on the device. At the 1980-2100 MT/s data rates, the device's DFE is disabled. Setting either the global DFE gain and tap 1-4 enable bits to disable or setting all DFE gain and tap 1-4 bias bits to step 0 disables the DFE. Ex. CC-04 (DDR5 Datasheet) at 263.
[1.g] control circuitry configured to receive the address and control signals from the system memory controller and to control the plurality of memory devices in response to the address and control signals,	The '731 Accused Products include control circuitry configured to receive the address and control signals from the system memory controller and to control the plurality of memory devices in response to the address and control signals. For example: The Micron's RDIMM products include a registering clock driver (RCD) that receives address and control ("CA") signals from the system memory and controls the plurality of memory devices in response to the address and control signals:

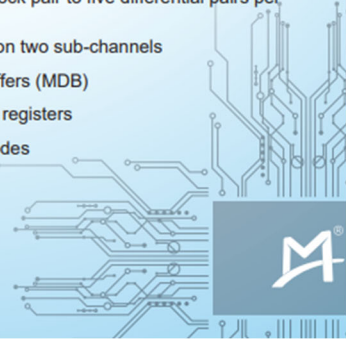
U.S. Patent No. 10,025,731 Infringement Claim Chart

Claim Language	'731 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (showing RCD receiving address and control signals (highlighted in yellow) and transmitting control signals to the memory banks).

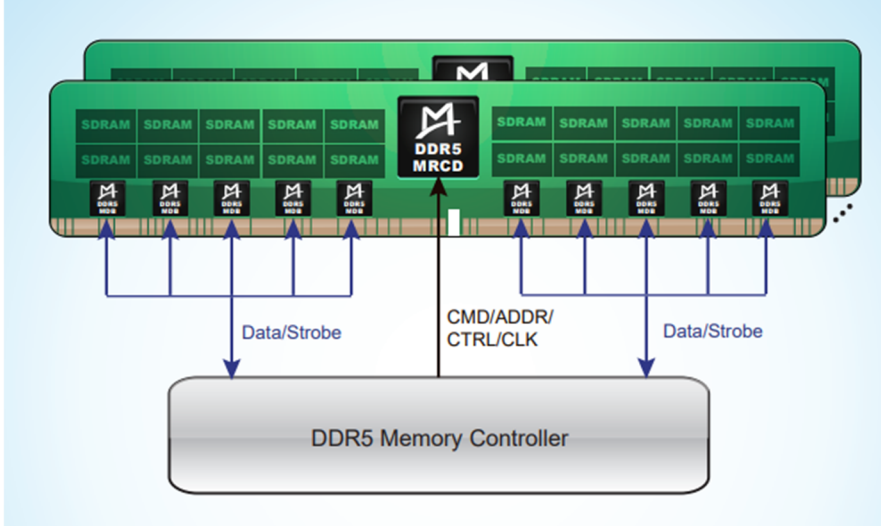
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	Table 4: Pin Descriptions <table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>CK_t, CK_c</td><td>Input</td><td>V_{DD}</td><td>Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>DDR Input</td><td>V_{DD}</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.</td></tr><tr><td>CS[1:0]_A CS[1:0]_B</td><td>Input</td><td>V_{DD}</td><td>Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V_{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DD}.</td></tr><tr><td>PAR_A PAR_B</td><td>Input</td><td>V_{DD}</td><td>Command and Address Parity Input: The RCD supports even parity check devices prior to forwarding commands to the DRAM. Once enabled on the RCD, the RCD calculates parity. Input parity should be maintained at the rising edge of the clock and at the same time with command and address with CSx_n LOW.</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 8. Micron’s MRDIMM products include a multiplexed rank registering clock driver (MRCD) that receives address and control (“CA”) signals from the system memory and controls the plurality of memory devices in response to the address and control signals:	Symbol	Type	I/O Level	Description	CK_t, CK_c	Input	V _{DD}	Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.	CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V _{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DD} .	PAR_A PAR_B	Input	V _{DD}	Command and Address Parity Input: The RCD supports even parity check devices prior to forwarding commands to the DRAM. Once enabled on the RCD, the RCD calculates parity. Input parity should be maintained at the rising edge of the clock and at the same time with command and address with CSx_n LOW.
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CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.																		
CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V _{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DD} .																		
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	General Description The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in applications such as deep learning and large-scale data analysis. Application • DDR5 MRDIMM / MCRDIMM Feature • Speed up to 8800MT/s • Two sub-channels, each divided into two pseudo-channels to increase the total bandwidth of the host system • Two pseudo-channels splitting the incoming CA inputs and generating separate CA outputs • Integrated PLL clock driver distributing one differential clock pair to five differential pairs per channel • Parity checking across CA and DPAR inputs separately on two sub-channels • Integrated BCOM bus to control the multiplexed data buffers (MDB) • Output characteristics configurable through control word registers • Sideband bus interface supporting I²C and I³C Basic modes • CA, CS, DFE and BCOM training modes support • Multiple power-saving modes • 1.1V VDD and 1.0V VDDIO voltages • Green package: 240-ball FBGA 

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	 Ex. CC-12 M88MR5RCD01 Flyer, Montage Technology. Analysis below uses RCDs as an example, but MRCDs would operate in the same way. <i>See</i> Ex. CC-13 JEDEC Press Release (planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor & Pinout, SPD, PMIC, and TS for ease of adoption”).
[1.h] wherein the control circuitry is further configured to dynamically control the at least one programmable impedance matching circuit in the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit based on which of the multiple ranks is selected	In the '731 Accused Products, the control circuitry is further configured to dynamically control the at least one programmable impedance matching circuit in the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit based on which of the multiple ranks is selected to communicate data with the system memory controller in response to the address and control signals. For example: In DDR5 systems, the control circuitry (e.g., RDIMM, MRDIMM) sends a ODT control commands encoded into the CS signal, and each rank of the memory module receives a separate Chip Select Command. For example, the RCD together with decoding and control circuitry on the SDRAMs controls the operation of the SDRAMs, including affecting the value of the RTT for the specific rank <i>See, e.g.:</i>

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to communicate data with the system memory controller in response to the address and control signals.	3.3.1 Chip Select Operation The component has two chip select inputs per channel, DCS0_n and DCS1_n, and two copies of each chip select output, QCS0_A_n, QCS1_A_n, QCS0_B_n and QCS1_B_n. Commands are sent to a single rank or multiple ranks, as determined by the DCS[1:0]_n inputs. The number of input chip selects matches the number of output chip selects in each of the two sets (A-outputs and B-outputs). Table 3 — DCS - QxCS Mapping <table><tr><th>Input CS</th><th>Output CS</th></tr><tr><td>DCS0_N_n</td><td>QxCS0_N_n</td></tr><tr><td>DCS1_N_n</td><td>QxCS1_N_n</td></tr></table> 3.3.2 ODT Operation DDR5 does not have a separate ODT (On Die Termination) signal going to the DIMM or the DRAMs. The ODT control is encoded into the chip selects. All commands that utilize the data bus require 2UI on the DCA bus in order to transfer the appropriate number of command and addressing bits. This requirement is used to signal ODT control by how the chip selects are asserted for the two UI. Table 4 — ODT Encoding on the Chip Select <table><tr><th>DCS_n on 1st UI</th><th>DCS_n on 2nd UI</th><th>QxCS[1:0]_n</th><th>DRAM DQ Terminations</th></tr><tr><td>Active</td><td>Inactive</td><td>Target of command</td><td>Read: none (device drives data) Write: RTT_WR</td></tr><tr><td>Active</td><td>Active</td><td>Non-Target Termination should be applied.</td><td>Read: RTT_NOM_RD Write: RTT_NOM_WR</td></tr><tr><td>Inactive</td><td>Inactive</td><td>DRAM ignores commands and the Parked termination will be maintained</td><td>RTT_PARK</td></tr><tr><td>Inactive</td><td>Active</td><td>Illegal</td><td>Illegal</td></tr></table> The host controller will determine the termination at the time of the command, by activating the chip selects appropriately. Ex. CC-32 (JESD82-512) at 8.	Input CS	Output CS	DCS0_N_n	QxCS0_N_n	DCS1_N_n	QxCS1_N_n	DCS_n on 1st UI	DCS_n on 2nd UI	QxCS[1:0]_n	DRAM DQ Terminations	Active	Inactive	Target of command	Read: none (device drives data) Write: RTT_WR	Active	Active	Non-Target Termination should be applied.	Read: RTT_NOM_RD Write: RTT_NOM_WR	Inactive	Inactive	DRAM ignores commands and the Parked termination will be maintained	RTT_PARK	Inactive	Active	Illegal	Illegal
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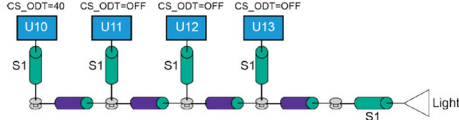
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	Description The RG5R256 (Gen 2 RCD) is a registering clock driver used on DDR5 RDIMMs, LRDIMMs and NVDIMMs. It supports DDR5 server speeds up to 5600 MT/s. Its primary function is to buffer the Command Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus to control the data buffers for LRDIMMs. The RG5R256 contains two separate channels with some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip-select inputs, produces two copies of 14-bit single data rate CA bus outputs and two copies of the chip-select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-51 at 1 (Renesas Gen 2 RCD Website); <i>see also</i> Ex. CC-52 at 1 (Renesas Gen 1 RCD Website); Ex. CC-99 at 1 (Renesas Gen 3 RCD Website); Ex. CC-100 at 2 (Renesas Gen 4 RCD Website). Feature List - Compliant with JEDEC DDR5RCD02 specification - Speed up to 5600MT/s 1.1V VDD and 1.0V VDDIO voltages - Two-channel 1:2 registering buffer for Command/Address inputs and control signals - Integrated PLL clock driver distributing one differential clock pair to five - Dual chip selects - Parity checking across CA and DPAR separately on two sub-channels - Output characteristics configurable through control registers - CA, CS and DFE training modes - Sideband bus interface supporting I²C and I3C Basic modes - Power saving modes: PDE power down, Self refresh w/wo Clock Stop modes - Green package: 240-ball FCBGA

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	Ex. CC-43 at 1 (Montage Tech RCD Brochure); <i>see also</i> Ex. CC-97 (same); Ex. CC-98 (same). 3.3 Register Operation 3.3.1 Chip Select Operation The component has two chip select inputs per channel, DCS0_n and DCS1_n, and two copies of each chip select output, QCS0_A_n, QCS1_A_n, QCS0_B_n and QCS1_B_n. Commands are sent to a single rank or multiple ranks, as determined by the DCS[1:0]_n inputs. The number of input chip selects matches the number of output chip selects in each of the two sets (A-outputs and B-outputs). Table 3 — DCS - QxCS Mapping <table><tr><th>Input CS</th><th>Output CS</th></tr><tr><td>DCS0_N_n</td><td>QxCS0_N_n</td></tr><tr><td>DCS1_N_n</td><td>QxCS1_N_n</td></tr></table> 3.3.2 ODT Operation DDR5 does not have a separate ODT (On Die Termination) signal going to the DIMM or the DRAMs. The ODT control is encoded into the chip selects. All commands that utilize the data bus require 2UI on the DCA bus in order to transfer the appropriate number of command and addressing bits. This requirement is used to signal ODT control by how the chip selects are asserted for the two UI. Table 4 — ODT encoding on the chip select <table><tr><th>DCS_n on 1st UI</th><th>DCS_n on 2nd UI</th><th>QxCS[1:0]_n</th><th>DRAM DQ Terminations</th></tr><tr><td>Active</td><td>Inactive</td><td>Target of command</td><td>Read: none (device drives data) Write: RTT_WR</td></tr><tr><td>Active</td><td>Active</td><td>Non-Target Termination should be applied.</td><td>Read: RTT_NOM_RD Write: RTT_NOM_WR</td></tr><tr><td>Inactive</td><td>Inactive</td><td>DRAM ignores commands and the Parked termination will be maintained</td><td>RTT_PARK</td></tr><tr><td>Inactive</td><td>Active</td><td>Illegal</td><td>Illegal</td></tr></table> The host controller will determine the termination at the time of the command, by activating the chip selects appropriately. Ex. CC-32 (JESD82-512) at 8 (annotated). In the Accused Products, the control circuitry (i.e., RCD in RDIMMs and MRCD in MRDIMMs) dynamically controls the programmable impedance in the ODT circuitry. For example, the Accused Products include a “CA_ODT pin [that] enables the distinction of two different ‘sets’ of CA/CS/CK ODT settings” including a “weak termination setting” and a “stronger termination setting.”:	Input CS	Output CS	DCS0_N_n	QxCS0_N_n	DCS1_N_n	QxCS1_N_n	DCS_n on 1st UI	DCS_n on 2nd UI	QxCS[1:0]_n	DRAM DQ Terminations	Active	Inactive	Target of command	Read: none (device drives data) Write: RTT_WR	Active	Active	Non-Target Termination should be applied.	Read: RTT_NOM_RD Write: RTT_NOM_WR	Inactive	Inactive	DRAM ignores commands and the Parked termination will be maintained	RTT_PARK	Inactive	Active	Illegal	Illegal
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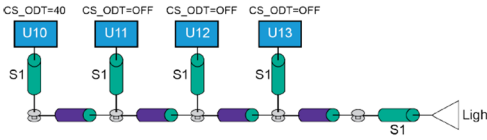
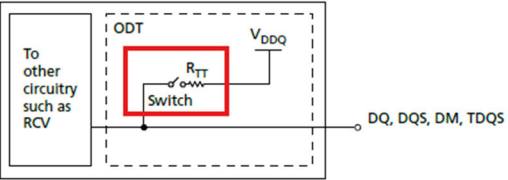
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	DDR5 On-Die Termination DDR5 module designs incorporate the same basic routing topologies for all I/O, address, control/command and clock signals as DDR4: • The familiar input/output (DQ) and input/output strobe (DQS) pins are all direct-routed from the edge connector or data buffer. • Clock, command and address pins are fly-by routed. The difference is with the termination method: DDR5 uses on-die termination (ODT) instead of discrete termination used on DDR4. • DDR4 uses discrete termination resistors on the modules/boards for command clock (CK), chip select (CS), command/address (CA) and other control pins. • DDR5 adds the benefit of programmable ODT for CK, CS and CA, as well as a per-device configurable CA_ODT pin. The CA_ODT pin is a new feature on each DDR5 SDRAM device that enables the last DRAM on a CS, CA, or CK net to have a comparatively strong ODT setting (40/80 ohms), while all remaining DRAM on the CS, CA, or CK net have weak or disabled (ODT) settings (disabled/480 ohms) (Figure 5). The CA_ODT pin can be tied/strapped HIGH or LOW on the board/DIMM to enable one of two different groupings, which have mode register-programmable ODT levels. Typical DIMM or board usage applies a weak termination setting to Group A (devices nearer to the controller) and a stronger termination setting to the Group B (device at the end of the fly-by routing). DDR5 CS_n Fly-by Routing and ODT Setting Example  Figure 5: DDR5 Additional Routing Technologies Ex. CC-11 (DDR5 Features White Paper) at 4. The Accused Products comply with the JEDEC JESD79-5 (<i>see</i> limitation [1.a] above), which further shows that the Accused Products include ODT circuitry and DFE circuitry that is coupled between the at least one connector and the plurality of memory devices.

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	On-Die Termination On-Die Termination for DQ, DQS, DM and TDS On-die termination (ODT) enables the DRAM to change termination resistance for each DQ, DQS, DM and TDQS pin. Unlike previous DDR technologies, DDR5 no longer has a physical ODT pin, and all ODT-based control is now command- and mode register-based. For x4 and x8 configuration devices, ODT is applied to each DQ, DQS_t, DQS_c and DM_n, as well as TDQS_t, TDQS_c for x8 configuration). For x16 configuration devices, ODT is applied to each DQU, DQL, DQSU_t, DQSU_c, DQSL_t, DQSL_c, DMU_n and DML_n. ODT is enabled via non-target READ/MODE REGISTER READ commands, WRITE and NON-TARGET WRITE commands, or is set to the default PARK value through settings in MR33-MR35. The ODT feature is designed to improve signal integrity of the memory channel by enabling the device controller to independently change termination resistance for any or all devices. In addition to the control capability of the DQ ODT, the DQS ODT is now independently programmed via MR33:OP[5:3] and held static. All ODT control is targeted for the DQs. This addition enables adjusting the delay common in an unmatched architecture. DQS RTT offset control mode is enabled via MR40:OP[2:0]. The ODT feature is turned off and not supported in self refresh mode, but does have an optional mode when in power-down. The figure below shows a simple functional representation of the ODT feature. Ex. CC-04 (DDR5 Datasheet) at 303. 4.40 CA_ODT Strap Operation With the introduction of on-die termination for CA/CS/CK on DDR5 DRAMs, the setting of the termination values per DRAM will be different depending on the configuration of DRAMs on the DIMM or system board. The CA_ODT pin enables the distinction of two "sets" of CA/CS/CK ODT settings. When the CA_ODT pin is strapped to a constant VSS setting on the DIMM or board, the CA/CS/CK ODT settings will be referred to as "Group A". When the CA_ODT pin is strapped to a constant VDD setting on the DIMM or system board, the CA/CS/CK ODT settings will be referred to as "Group B". Typical usage would be to apply a weak termination setting to Group A devices and a stronger termination setting to Group B devices, which would be at the end of the fly-by routing on the DIMM. To support these different settings, two sets of MPC opcodes will be used to target either Group A or Group B devices. In addition to these separate "Groups" of devices based on the CA_ODT pin, the PDA commands will also be supported. However, the correct combination of the PDA Select ID and MPC opcode must be used according to the CA_ODT pin value. For DDP device, CA_ODT package pin ties to both top die and bottom die CA_ODT bond pads. CS_n and CS1_n being defined in DDP package ballout. CS is not shared between dies, and each die terminated individually. CK is shared between both dies and both dies terminated. CA is shared between both dies and both dies terminated. host can program optimal ODT value after initialization. The CA_ODT pin is defined in the pinlist as follows:

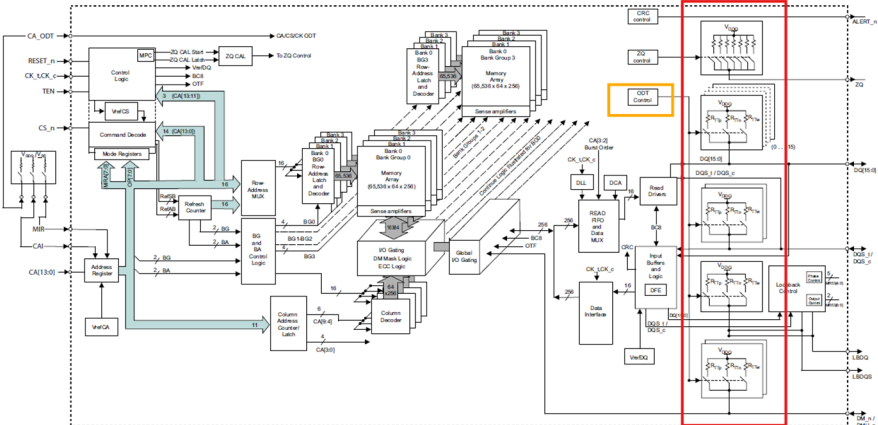
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	Ex. CC-05 (JESD79-5) at 299. DDR5 On-Die Termination DDR5 module designs incorporate the same basic routing topologies for all I/O, address, control/command and clock signals as DDR4: • The familiar input/output (DQ) and input/output strobe (DQS) pins are all direct-routed from the edge connector or data buffer. • Clock, command and address pins are fly-by routed. The difference is with the termination method: DDR5 uses on-die termination (ODT) instead of discrete termination used on DDR4. • DDR4 uses discrete termination resistors on the modules/boards for command clock (CK), chip select (CS), command/address (CA) and other control pins. • DDR5 adds the benefit of programmable ODT for CK, CS and CA, as well as a per-device configurable CA_ODT pin. The CA_ODT pin is a new feature on each DDR5 SDRAM device that enables the last DRAM on a CS, CA, or CK net to have a comparatively strong ODT setting (40/80 ohms), while all remaining DRAM on the CS, CA, or CK net have weak or disabled (ODT) settings (disabled/480 ohms) (Figure 5). The CA_ODT pin can be tied/strapped HIGH or LOW on the board/DIMM to enable one of two different groupings, which have mode register-programmable ODT levels. Typical DIMM or board usage applies a weak termination setting to Group A (devices nearer to the controller) and a stronger termination setting to the Group B (device at the end of the fly-by routing). DDR5 CS_n Fly-by Routing and ODT Setting Example  Figure 5: DDR5 Additional Routing Technologies Ex. CC-11 (DDR5 Features White Paper) at 4. Figure 192: Functional Representation of ODT  The switch is enabled by the internal ODT control logic, which uses command decode, mode register settings and other control information (see below). The value of R_{TT} is determined by the settings of mode register bits.

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	Ex. CC-04 (DDR5 Datasheet) at 303 (red box shows the programmable switch that dynamically places a termination resistor for impedance matching). Table 182 — ODT Electrical Characteristics RZQ=240 Ω +/-1% Entire Temperature Operation Range; after Proper ZQ Calibration <table><tr><th>RTT</th><th>Vout</th><th>Min</th><th>Nom</th><th>Max</th><th>Unit</th><th>Notes</th></tr><tr><td rowspan="3">240 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ</td><td>1, 2, 3</td></tr><tr><td rowspan="3">120 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/2</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/2</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/2</td><td>1, 2, 3</td></tr><tr><td rowspan="3">80 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/3</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/3</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/3</td><td>1, 2, 3</td></tr><tr><td rowspan="3">60 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/4</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/4</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/4</td><td>1, 2, 3</td></tr><tr><td rowspan="3">48 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/5</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/5</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/5</td><td>1, 2, 3</td></tr><tr><td rowspan="3">40 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/6</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/6</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/6</td><td>1, 2, 3</td></tr><tr><td rowspan="3">34 Ω</td><td>VOLdc= 0.5* VDDQ</td><td>0.9</td><td>1</td><td>1.25</td><td>RZQ/7</td><td>1, 2, 3</td></tr><tr><td>VOMdc= 0.8* VDDQ</td><td>0.9</td><td>1</td><td>1.1</td><td>RZQ/7</td><td>1, 2, 3</td></tr><tr><td>VOHdc= 0.95* VDDQ</td><td>0.8</td><td>1</td><td>1.1</td><td>RZQ/7</td><td>1, 2, 3</td></tr><tr><td>DQ-DQ Mismatch within byte</td><td>VOMdc = 0.8* VDDQ</td><td>0</td><td>-</td><td>8</td><td>%</td><td>1, 2, 4, 5, 6</td></tr></table> NOTE 1 The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see "ZQ Calibration Commands" section. NOTE 2 Pull-up ODT resistors are recommended to be calibrated at 0.8*VDDQ. Other calibration schemes may be used to achieve the linearity spec shown above, e.g. calibration at 0.5*VDDQ and 0.95*VDDQ. NOTE 3 The tolerance limits are specified under the condition that VDDQ=VDD. NOTE 4 DQ to DQ mismatch within byte variation for a given component including DQS_T and DQS_C (characterized) NOTE 5 RTT variance range ratio to RTT Nominal value in a given component, including DQS_t and DQS_c. $\text{DQ-DQ Mismatch in a Device} = \frac{\text{RTTMax} - \text{RTTMin}}{\text{RTTNOM}} * 100$ NOTE 6 This parameter of x16 device is specified for Upper byte and Lower byte. Ex. CC-05 (JESD79-5) at 315 (showing the electrical characteristics for the resistance values that can be selected for a termination (RTT) for impedance matching); <i>see also</i> Ex. CC-04 (DDR5 Datasheet) at 304 (same).	RTT	Vout	Min	Nom	Max	Unit	Notes	240 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ	1, 2, 3	120 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/2	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/2	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/2	1, 2, 3	80 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/3	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/3	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/3	1, 2, 3	60 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/4	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/4	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/4	1, 2, 3	48 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/5	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/5	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/5	1, 2, 3	40 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/6	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/6	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/6	1, 2, 3	34 Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/7	1, 2, 3	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/7	1, 2, 3	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/7	1, 2, 3	DQ-DQ Mismatch within byte	VOMdc = 0.8* VDDQ	0	-	8	%	1, 2, 4, 5, 6
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Claim Language	'731 Accused Products ⁴
	Figure 4: 1 Gig x16 Functional Block Diagram  Ex. CC-09 16Gb DDR5 SDRAM Addendum at 6 (showing ODT control, in orange, dynamically controlling programmable impedance matching in the ODT circuitry); <i>see also id.</i> at Figures 2, 4. The Accused Products also include “Dynamic ODT” to “further enhance signal integrity on the data bus.”

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	5.3 Dynamic ODT In certain application cases and to further enhance signal integrity on the data bus, it is desirable that the termination strength of the DDR5 SDRAM can be changed without issuing an MRW command. This requirement is supported by the "Dynamic ODT" feature as described as follows: 5.3.1 ODT Functional Description The function is described as follows: • Five RTT values are available: RTT_NOM_RD, RTT_NOM_WR, RTT_PARK, RTT_WR and DQS_RTT_PARK. - The value for RTT_NOM_RD is preselected via MR35:OP[5:3] - The value for RTT_NOM_WR is preselected via MR35:OP[2:0] - The value for RTT_WR is preselected via MR34:OP[5:3] - The value for RTT_PARK is preselected via MR34:OP[2:0] - Programmed via MPC - The value for DQS_RTT_PARK is preselected via MR33:OP[5:3] - Programmed via MPC • During operation without commands, the termination is controlled as follows: - Nominal termination strength for all types (RTT_NOM_RD, RTT_NOM_WR, RTT_WR, RTT_PARK & DQS_RTT_PARK) are selected. - RTT_NOM_RD and RTT_NOM_WR on/off timings are controlled via the respective non-target Read/Mode Register Read and Write commands and latencies. - DQS_RTT_PARK is held static and is based on the value programmed in the MR listed above. • When a Write command (WR) is registered, the termination is controlled as follows: - A latency ODTLon_WR after the Write command, termination strength RTT_WR is selected. - A latency ODTOff_WR after the Write command, termination strength RTT_WR is de-selected.
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	- The termination, RTT_NOM_WR, for the non-target Write command is selected and de-selected by latencies ODTLon_WR_NT and ODTLoFF_WR_NT, respectively. - When a Read command (RD) is registered, the termination is controlled as follows: - A latency ODTLoFF_RD after the Read command, data termination is disabled. Then, ODTLon_RD after the Read command, data termination is enabled. - A latency ODTLoFF_RD_DQS after the Read command, strobe termination is disabled. ODTLon_RD_DQS after the Read command, strobe termination is enabled. - The termination, RTT_NOM_RD, for the non-target Read command is selected and de-selected by latencies ODTLon_RD_NT and ODTLoFF_RD_NT, respectively. The duration of a Write or Read command is a full burst cycle, BL/2. The termination select ("ODTLon,...") and de-select ("ODTLoFF,...") latency settings shall not result in an ODT pulse width which violates a burst cycle (BL/2) minimum duration. The equation "ODTLoFF_X - ODTLon_X >= BL/2" must be met, where X is the termination latency setting associated with a particular command type (WR, WR_NT, RD_NT). To achieve the minimum write burst duration, ODTLoFF_X and ODTLon_X latencies contain independent programmable mode register offsets: - The values for the Write command ODT control offsets are preselected via MR37. MR37:OP[2:0] preselects ODTLon_WR_Offset MR37:OP[5:3] preselects ODTLoFF_WR_Offset - The values for the non-target Write command ODT control offsets are preselected via MR38. MR38:OP[2:0] preselects ODTLon_WR_NT_Offset MR38:OP[5:3] preselects ODTLoFF_WR_NT_Offset - The values for the non-target Read command ODT control offsets are preselected via MR39. MR39:OP[2:0] preselects ODTLon_RD_NT_Offset MR39:OP[5:3] preselects ODTLoFF_RD_NT_Offset Ex. CC-05 (JESD79-5) at 316. For example, the control circuitry controls the programmable impedance based on which of the multiple ranks is selected (i.e., through input CS_n): <table><tr><td>CS[1:0]_A CS[1:0]_B</td><td>Input</td><td>V_{DD}</td><td> TER SET commands. Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V_{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DD}. Command and Address Parity Inputs: The BCD supports even parity check </td></tr></table>	CS[1:0]_A CS[1:0]_B	Input	V _{DD}	TER SET commands. Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V_{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DD}. Command and Address Parity Inputs: The BCD supports even parity check
CS[1:0]_A CS[1:0]_B	Input	V _{DD}	TER SET commands. Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V_{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DD}. Command and Address Parity Inputs: The BCD supports even parity check		

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	Ex. CC-03 (DDR5 RDIMM Datasheet) at 8. In the '731 Accused Products, the programmable impedance matching circuit in the ODT circuitry can be independently programmed based on whether a rank is active or not active: Application: Controller can control each RTT condition with the WRITE (WR), READ (RD), or MODE REGISTER READ (MRR) commands use of ODT offset control mode registers. • Data Termination Disable: Device driving data upon receiving READ command disables the termination after CL-1 and re-enables the termination at CL + BL/2 (CL and BL defined in clock cycles). • Strobe Termination Disable: Device driving strobe upon receiving READ command disables the termination after CL-1-'RPRE-(Read_DQS_Offset) and re-enables the termination at CL+BL/2-0.5+'RPST-(Read_DQS_Offset) (CL and BL defined in clock cycles). • RTT_WR: The rank that is being written to provides termination and adjusts timing based on the ODT control mode register settings. • RTT_NOM_RD: The device turns this on if it sees CS asserted during the second pulse of the READ command (except when ODT is disabled by MR35:OP[5:3]). • RTT_NOM_WR: The device turns this on if it sees a CS asserted during the second pulse of the WRITE command (except when ODT is disabled by MR35:OP[2:0]). • RTT_PARK: The default parked value (set via MR34[2:0]) to be enabled when a READ or WRITE is not active. • DQS_RTT_PARK: The default parked value (set for DQS via MR33:OP[5:3]); it is enabled when a READ or WRITE is not active. The RTT values have priority as follows: 1. Data termination disable and strobe termination disable 2. RTT_WR 3. RTT_NOM_RD 4. RTT_NOM_WR 5. RTT_PARK and DQS_RTT_PARK This means if there is WRITE command, the device turns on RTT_WR (not RTT_NOM_RD or RTT_NOM_WR). Additionally, if there is READ command, the device disables data termination and enters driving mode. If during the second cycle of a WRITE, READ or MODE REGISTER READ command the CS is held LOW, the NT ODT is enabled and the appropriate RTT_NOM_WR or RTT_NOM_RD is enabled for the non-target rank. This provides additional—and potentially different—term options for the other ranks on the channel. Ex. CC-04 (DDR5 Datasheet) at 305.

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	5.3 Dynamic ODT In certain application cases and to further enhance signal integrity on the data bus, it is desirable that the termination strength of the DDR5 SDRAM can be changed without issuing an MRW command. This requirement is supported by the "Dynamic ODT" feature as described as follows: 5.3.1 ODT Functional Description The function is described as follows: • Five RTT values are available: RTT_NOM_RD, RTT_NOM_WR, RTT_PARK, RTT_WR and DQS_RTT_PARK. - The value for RTT_NOM_RD is preselected via MR35:OP[5:3] - The value for RTT_NOM_WR is preselected via MR35:OP[2:0] - The value for RTT_WR is preselected via MR34:OP[5:3] - The value for RTT_PARK is preselected via MR34:OP[2:0] - Programmed via MPC - The value for DQS_RTT_PARK is preselected via MR33:OP[5:3] - Programmed via MPC • During operation without commands, the termination is controlled as follows: - Nominal termination strength for all types (RTT_NOM_RD, RTT_NOM_WR, RTT_WR, RTT_PARK & DQS_RTT_PARK) are selected. - RTT_NOM_RD and RTT_NOM_WR on/off timings are controlled via the respective non-target Read/Mode Register Read and Write commands and latencies. - DQS_RTT_PARK is held static and is based on the value programmed in the MR listed above. • When a Write command (WR) is registered, the termination is controlled as follows: - A latency ODTLon_WR after the Write command, termination strength RTT_WR is selected. - A latency ODTOff_WR after the Write command, termination strength RTT_WR is de-selected.
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	• The termination, RTT_NOM_WR, for the non-target Write command is selected and de-selected by latencies ODTLon_WR_NT and ODTLoFF_WR_NT, respectively. • When a Read command (RD) is registered, the termination is controlled as follows: - A latency ODTLoFF_RD after the Read command, data termination is disabled. Then, ODTLon_RD after the Read command, data termination is enabled. - A latency ODTLoFF_RD_DQS after the Read command, strobe termination is disabled. ODTLon_RD_DQS after the Read command, strobe termination is enabled. • The termination, RTT_NOM_RD, for the non-target Read command is selected and de-selected by latencies ODTLon_RD_NT and ODTLoFF_RD_NT, respectively. The duration of a Write or Read command is a full burst cycle, BL/2. The termination select ("ODTLon,...") and de-select ("ODTLoFF,...") latency settings shall not result in an ODT pulse width which violates a burst cycle (BL/2) minimum duration. The equation "ODTLoFF_X - ODTLon_X >= BL/2" must be met, where X is the termination latency setting associated with a particular command type (WR, WR_NT, RD_NT). To achieve the minimum write burst duration, ODTLoFF_X and ODTLon_X latencies contain independent programmable mode register offsets: - The values for the Write command ODT control offsets are preselected via MR37. - MR37:OP[2:0] preselects ODTLon_WR_Offset - MR37:OP[5:3] preselects ODTLoFF_WR_Offset - The values for the non-target Write command ODT control offsets are preselected via MR38. - MR38:OP[2:0] preselects ODTLon_WR_NT_Offset - MR38:OP[5:3] preselects ODTLoFF_WR_NT_Offset - The values for the non-target Read command ODT control offsets are preselected via MR39. - MR39:OP[2:0] preselects ODTLon_RD_NT_Offset - MR39:OP[5:3] preselects ODTLoFF_RD_NT_Offset Ex. CC-05 (JESD79-5) at 316 (describing operation of ODT). 5.3.3 ODT Timing Diagrams The following pages provide examples of ODT utilization timing diagrams. Examples of Write to Write, Read to Write and Read to Read are provided for clarification only. Implementations may vary, including termination on other DIMMS. It is the controller's responsibility to manage command spacing and the programmable aspect of tODLon/off times to ensure that preambles and postambles are included in the RTT ON time. When there is a 1 tCK ODT control gap for any ODT operation (such as shown in Figure 199), the said gap's RTT value will be the same or smaller (stronger termination) than RTT_PARK. All timings noted in the following figures are just used as reference. Ex. CC-05 (JESD79-5) at 321.

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